

# Eli Whitaker

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## Summary

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Mechanical Engineering graduate from UCLA with a 4.00 GPA and hands-on experience building vision-guided robotic systems. Strong in perception-to-action pipelines, calibration, motion control, embedded actuation, and debugging real electromechanical prototypes.

## Education

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### University of California, Los Angeles

Los Angeles, CA

*B.S. Mechanical Engineering | GPA: 4.00 | Summa Cum Laude*

*June 2026*

- Relevant coursework: Robotics, Mechanical Design, Manufacturing Processes, Dynamic Systems, Controls, Materials Science, Fluid Mechanics, Thermodynamics, Statics and Strength of Materials.

## Technical Skills

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- **Robotics / Controls:** kinematics, inverse kinematics, coordinate transforms, visual servoing, PD/PID control, inverse dynamics, trajectory execution, state machines, failure recovery.
- **Perception / Autonomy:** YOLO segmentation, stereo vision, RAFT-Stereo, OpenCV, masked point clouds, homography calibration, target re-identification, 2.5D planning.
- **Software / Hardware:** Python, C++, MATLAB, ROS 2, Git/GitHub, Teensy 4.1, Jetson, stepper motors, TMC2209 drivers, Dynamixel actuators, servo grippers, cameras.
- **Mechanical / Prototyping:** SolidWorks, Onshape, mechanism layout, actuator packaging, CAD-to-build iteration, 3D printing, benchtop testing, electromechanical debugging.

## Research Experience

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### UCLA Structures-Computer Interaction Lab

Los Angeles, CA

*Undergraduate Researcher, Robotic Laser Weeding*

*Aug. 2025 – Jun. 2026*

- Developed autonomy components for a robotic diode-laser weeding platform integrating a 60 W blue diode laser, eye-in-hand stereo vision, CNC gantry motion, Jetson edge compute, target re-identification, and visual servoing.
- Built perception and control workflows for burst-stable weed detection, stereo triangulation, route planning, coarse gantry positioning, local target re-identification, PD alignment, and timed laser firing.
- Supported system characterization across 70 autonomous trials and 525 planned treatments, decomposing runtime into travel, re-identification, fine alignment, and laser dwell bottlenecks.

## Engineering Projects

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### Autonomous Grocery Bagging Robot

*UCLA MAE 162D/E Capstone*

- Led autonomy stack for a closed-loop grocery packing robot using YOLO segmentation, RAFT-Stereo disparity, masked point-cloud reconstruction, height-indexed overhead homographies, and camera-to-robot calibration.
- Converted camera detections into robot-frame pick/place targets for a SCARA-like RRPR manipulator with stepper-driven joints, vertical prismatic motion, servo gripper control, and Teensy 4.1 low-level execution.
- Implemented deterministic 2.5D AABB packing logic that evaluated footprint fit, collision/overlap, support, overhang, usable bag volume, and top-down descent feasibility before each placement.
- Built missed-pick recovery behavior using post-grasp YOLO continuity checks; achieved 95.1% pick-and-place success (404/425 attempts) across 64 autonomous runs on physical hardware.

### Vision-Guided Ball-Catching Robot

*UCLA Robotics*

- Designed and tested operational-space inverse dynamics control for a planar 2R Dynamixel manipulator intercepting ping-pong balls in a 0.30 m by 0.30 m workspace.
- Integrated a ROS 2 vision/control architecture using high-speed stereo tracking, OpenCV contour detection, IIR-filtered trajectory prediction, encoder feedback, and real-time actuator commands.

## Additional Experience

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### UCLA Recreation / John Wooden Center

Los Angeles, CA

*FITWELL Consultant / Supervisor*

*Summer 2025 – Jun. 2026*

- Supported day-to-day facility operations, equipment issue tracking, staff coordination, and patron communication in a high-traffic university recreation environment.

### Casa Roble High School

Rancho Cordova, CA

*Volunteer Math Tutor*

*Summer 2023*

- Tutored AP Calculus and Integrated Math students, adapting technical explanations to student needs.